

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1

Mark scheme

AQA spec 3.2.4; The physical environment; textbook pp. 148-157

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• 7×10^{11} tonnes	1
01.2	• 10% of 10 000 = 1000 billion tonnes • $1000 / 700 \times 100 = 143\%$ (accept 142.9)	2
01.3	• Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) • Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion	2
01.4	• CO₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis • Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons • When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods	3
01.5	• Aerobic respiration uses oxygen to break down organic compounds completely to CO₂ (and water), releasing much more energy • Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH₄), which is later oxidised to CO₂ in the atmosphere	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	

Question	Marking guidance	Marks
02.1	• Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises • Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) • Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises	6
02.2	<i>Any 3 from:</i> • Rice paddy fields • Landfill sites • Anaerobic sediments in reservoirs • Livestock intestines (cattle) • Manure / slurry stores	3
02.3	• More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
03.1	• Trees absorb CO₂ by photosynthesis and store carbon in wood as they grow • Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir • Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow	3
03.2	• Pre-combustion: the fuel is modified before use so no CO₂ is released when burnt - eg gasification of coal to hydrogen and CO₂ (CO₂ stored, hydrogen distributed); oxy-fuel combustion in pure oxygen giving only CO₂ and water vapour which is condensed out • Advantage: capture at a large central facility; suits small dispersed sources (vehicles) that cannot capture CO₂ • Post-combustion: CO₂ removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene • Disadvantage: only about 20% of flue gas is CO₂ so separation is costly / energy-intensive; none fully developed	4
03.3	<i>Any 3 from:</i> • Injected into depleted oil / gas fields or saline aquifers • Injected into oil reservoirs to provide pressure for secondary oil recovery • Reaction with fractured basalt to form solid carbonate minerals • Concern: leakage back to the atmosphere / monitoring over centuries; limited suitable geology; cost	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
04.1	• 1 nitrogen fixation; 2 nitrification • 3 denitrification; 4 ammonification • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three)	3

Question	Marking guidance	Marks
04.2	• Drainage makes the soil aerobic • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas	3
04.3	• Nitrogen gas is very unreactive so most organisms cannot use it directly • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount	2
04.4	• Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
05.1	• Human = $120 + 60 + 30 = 210$; total = 325 • $210 / 325 \times 100 = 64.6\%$ (accept 65%)	2
05.2	• Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure; ammonia then converted to nitrates for fertiliser • Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO₂ and NO_x • Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy)	3
05.3	<i>Any 3 from:</i> • Eutrophication of water bodies by leached nitrates / sewage • Photochemical smogs and tropospheric ozone from NO_x • Acid rain (nitric acid) • Nitrous oxide as a greenhouse gas • NO_x toxicity / respiratory disease • Nitrate in drinking water (methaemoglobinaemia)	3
05.4	• Carbon is lost as CO₂ (composting) or CH₄ (digestion) so the C:N ratio improves / more nitrogen-rich • Bulk and odour reduced, pathogens killed, easier to apply; digestion also yields biogas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
06.1	• There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) • Most phosphorus compounds have low solubility so little is available in soil water • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks	3
06.2	• Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates	2

Question	Marking guidance	Marks
06.3	• Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland	3
06.4	<i>Any 2 from:</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock • Breed crops that absorb phosphate more efficiently • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter - Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity - Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) - Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones - Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
07.2	The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	
08.1	$(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$	1
08.2	- Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ - Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter	2
08.3	<i>Any 2 from:</i> - Low / zero tillage - Adding manure, compost or mulch - Growing cover crops / leaving crop residues - Converting to long-term grass or tree crops in rotation	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	